Autonomous Parafoil Targeted Landing

Guidance Algorithm, Flight Firmware and Verification Report

Energy-managed guidance for a 1 kg CanSat under a steerable single-skin parafoil, with an 80 m landing-accuracy requirement

Team:	CU Jammu Astro, CanSat Team 039
Programme:	ISRO / IN-SPACe CanSat India Student Competition 2026
Subsystem:	Recovery, Guidance and Flight Software
Flight computer:	Teensy 4.1 (ARM Cortex-M7, 600 MHz)
Requirement:	Touchdown within 80 m of the target coordinate
Status:	Bench and ground tests passed; not yet drop-tested
Date:	5 October 2026

This report documents the complete guided-landing system as built: the problem analysis, every estimator and control law with its mathematics, the flight-state machine, the firmware implementation, the Monte-Carlo verification of the exact flight code, and the hardware bench results to date. It states plainly what has been verified and what has not.

Firmware package: `Parafoil_Guided_Landing/` (`Parafoil_Guided_Landing.ino`, `ParafoilGuidance.h`, `ResumeRecord.h`, `host_sim/monte_carlo.cpp`)

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1 Executive Summary

The CanSat descends under a drogue to $500\,\mathrm{m}$ above ground, then releases a steerable single-skin parafoil. The task is to land within $80\,\mathrm{m}$ of a target coordinate.

The earlier controller pointed the canopy at the target and centred the servos inside a $35\,\mathrm{m}$ "capture" circle. A parafoil cannot stop or slow down; it only glides. Pointing at the target makes it arrive with height to spare and land wherever the leftover glide takes it. The team's own simulation of that logic produced a $617.7\,\mathrm{m}$ miss.

The new system manages *energy*, not just direction:

1. It computes a **virtual target**: the point in the moving air that the wind will carry onto the real target exactly at touchdown. This turns the windy problem into a still-air problem.
2. It compares the glide still available with the distance to that point. The difference is **excess energy**, measured in metres of air path.
3. It flies four phases: **HOMING** toward the virtual target, **LOITER** in a $50\,\mathrm{m}$ orbit to burn excess height, **FINAL** approach with S-turns to burn the last metres, and **TERMINAL** wings-level flight below $8\,\mathrm{m}$.
4. Wind and airspeed are **measured in flight** from GPS ground velocity during turns; sink rate is measured by the barometer. No canopy parameter needs to be known in advance.

Verification. The guidance core is a single C++ header compiled identically by the Teensy firmware and by a PC Monte-Carlo harness. Over 3000 randomised flights (canopy, wind, sensors, GPS dropouts all varied):

Flight class	Flights	Within 80 m	Median miss
All physically reachable targets	2684	**98.3 %**	$14\,\mathrm{m}$
Reach margin above $150\,\mathrm{m}$	2451	**99.8 %**	$13.9\,\mathrm{m}$ (95th pct. $34\,\mathrm{m}$)
Old point-at-target logic, same reachable flights	2684	97.2 %	$30\,\mathrm{m}$

Hardware status. All sensors, both steering servos, the GPS at $5\,\mathrm{Hz}$, the in-firmware simulated flight ($13.9\,\mathrm{m}$ miss on real servos) and an outdoor walking test have passed. SD logging and the hardware watchdog are the remaining bench items. **No drop test has been flown yet**; every canopy number used so far is a modelling range, not a measurement (Section 12).

2 Problem Definition

2.1 Mission profile

From the CanSat CDR: ejection at apogee (about $1000\,\mathrm{m}$ AGL), a conical drogue to stabilise the payload at roughly $20\,\mathrm{m/s}$, servo-actuated release of the parafoil at $500\,\mathrm{m}$ AGL, an "8-second patience" window for pendulum damping, then guided descent. Steering is by pulling the left or right trailing-edge brake line through a continuous-rotation ST-3009 servo and spool.

2.2 Why pointing at the target fails

Let h be height above the target ground, V_z the sink rate and V_a the airspeed. In still air, the horizontal distance the canopy can still fly is

$$L = V_a\, t_{go}, \qquad t_{go} = \frac{h}{V_z}. \tag{1}$$

With $V_a = 6\,\mathrm{m/s}$ and $V_z = 2.5\,\mathrm{m/s}$, $L \approx 1200\,\mathrm{m}$ from $500\,\mathrm{m}$. If the target is $400\,\mathrm{m}$ away, flying straight at it leaves about $800\,\mathrm{m}$ of glide unused on arrival. A "capture" rule that stops steering at that point lets the canopy glide straight on and land hundreds of metres beyond. Accuracy therefore depends on *using up the right amount of glide at the right place*, which is an energy-management problem.

2.3 Reachability

Wind $\mathbf{W}$ moves the whole air mass. Over the remaining time the canopy drifts by $\mathbf{W}t_{go}$, and it can additionally cover any displacement of length up to $V_a t_{go}$ relative to the air. A target at the origin is reachable from position $\mathbf{p}$ only if

$$\left\| -\mathbf{p} - \mathbf{W}\, t_{go} \right\| \leq V_a\, t_{go}. \tag{2}$$

No guidance law can beat this bound. If the deployment point is outside the reachable region, the best possible result is the minimum-distance landing, and the 80 m requirement cannot be met regardless of software.

3 System Architecture

3.1 Hardware used for the guided-landing system

Function	Device	Interface and notes
Flight computer	Teensy 4.1	600 MHz Cortex-M7 with double-precision FPU
GNSS	Foxeer M10Q-180 (u-blox M10)	UART Serial3, 38 400 baud; GPS TX → pin 15 (RX3), GPS RX → pin 14 (TX3); VCC 5 V
Barometer	Bosch BMP581	I^{2}C 0x46/0x47, register-level driver
IMU	LSM6DSO32	I^{2}C 0x6A; yaw-rate source
Magnetometer	QMC5883	I^{2}C 0x0D; heading fallback only
Steering servos	2 × Feetech ST-3009	Serial1 half-duplex, 1 Mbps, IDs 1 (left) and 2 (right), neutral 2048
Deployment servo	SG90	PWM pin 8; 140° locked, 180° released
Data logger	Teensy built-in microSD	`FLTnnn.CSV` at 10 Hz

The I^{2}C bus is shared on SDA pin 18 and SCL pin 19 at 400 kHz. Sensors run from the Teensy 3.3 V rail; the ST-3009 servos have their own supply (measured 7.6 V) with a common ground.

3.2 Software structure

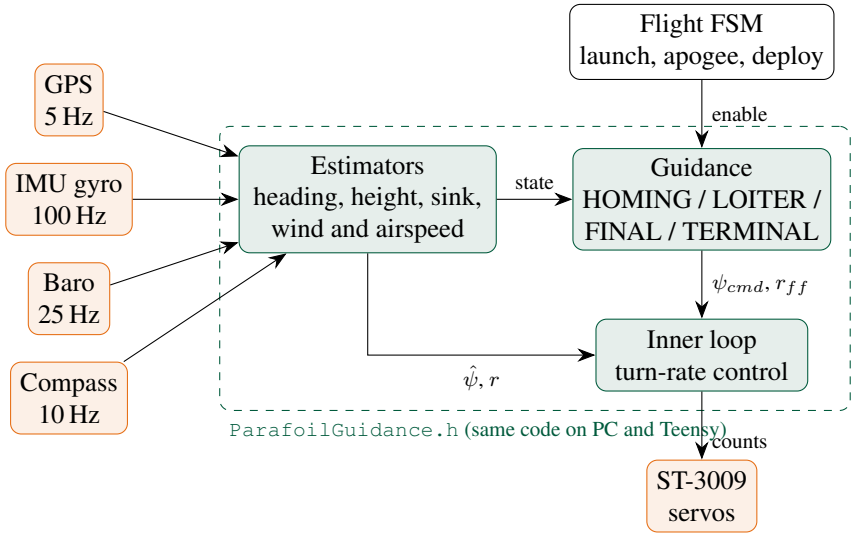


Figure 1: Data flow. Everything inside the dashed box lives in `ParafoilGuidance.h` as the class `FlightCore`. The `.ino` handles hardware, the flight-state machine, commands and logging.

File	Contents
`ParafoilGuidance.h`	Pure C++, no Arduino dependency: geodesy, filters, wind estimator, guidance, inner loop, `FlightCore`, simulation model.
`Parafoil_Guided_Landing.ino`	Sensor drivers, servo bus, flight-state machine, EEPROM resume, SD log, USB command interface, SIM and WALK bench modes.
`ResumeRecord.h`	EEPROM record for in-flight reset recovery (separate header so the Arduino prototype generator does not break it).
`host_sim/monte_carlo.cpp`	PC test harness: randomised flights through the same `FlightCore`.

4 Conventions and Coordinate Frames

- **Local frame:** North and East in metres, origin at the target.
- **Heading:** degrees; 0° is North, angles increase clockwise, so 90° is East. $\mathrm{wrap}_{180}(\cdot)$ maps an angle to $(-180^\circ, 180^\circ]$.
- **Steering demand u:** servo counts, **positive = turn right**. Saturated at ± 450 counts (about $\pm 40^\circ$ of servo rotation).
- **Height h:** metres above the target's ground level. **Sink rate** $V_z > 0$ means descending.
- **Wind W:** the direction the air is moving *to*.

4.1 Latitude and longitude to local metres

An equirectangular projection about the target (φ_0, λ_0):

$$m_\varphi = 111132.92 - 559.82\cos 2\varphi_0 + 1.175\cos 4\varphi_0 \quad [\mathrm{m}/^\circ], \tag{3}$$

$$m_\lambda = 111412.84\cos\varphi_0 - 93.5\cos 3\varphi_0 \quad [\mathrm{m}/^\circ], \tag{4}$$

$$p_N = (\varphi - \varphi_0)\, m_\varphi, \qquad p_E = (\lambda - \lambda_0)\, m_\lambda. \tag{5}$$

The error is below 0.1 % over a few kilometres, far smaller than GPS noise. Latitudes are kept in double precision.

5 State Estimation

5.1 Barometric height

At ARM the firmware averages 25 pressure samples to obtain the ground reference p_0. Height above the target is

$$h_{raw} = 44330\left[1 - \left(\frac{p}{p_0}\right)^{0.190295}\right] - \Delta h_{tgt} \quad [\mathrm{m}], \tag{6}$$

where Δh_{tgt} (`TARGET_ELEV_OFFSET_M`) is the target ground height minus the ARM-point height. If the barometer fails, GPS altitude minus its ARM value is used instead.

Alpha-beta filter. Height $\hat{h}$ and vertical speed $\hat{v}$ (positive up) are smoothed at 25 Hz with $\alpha = 0.12$, $\beta = 0.008$:

$$\hat{h}^- = \hat{h} + \hat{v}\,\Delta t, \qquad e = h_{raw} - \hat{h}^-, \tag{7}$$

$$\hat{h} = \hat{h}^- + \alpha\, e, \qquad \hat{v} = \hat{v} + \frac{\beta}{\Delta t}\, e. \tag{8}$$

A gap $\Delta t > 0.5$ s re-seeds the height and keeps the velocity.

5.2 Sink rate

Two first-order low-pass estimates with time constant $\tau_z = 3\,\mathrm{s}$ are kept:

- $V_{z,all}$, updated continuously after canopy opening. Its history is discarded at $0.6\,T_{damp}$ (4.8 s) to remove the opening shock.
- $V_{z,str}$, updated only after the yaw rate has stayed below $6\,^\circ/\mathrm{s}$ for 2 s (straight flight).

Guidance uses $V_{z,str}$ when available, otherwise $V_{z,all}$, otherwise the default $2.5\,\mathrm{m/s}$. The straight-flight value is preferred because the final approach is flown straight; turning increases sink and would bias t_{go}.

5.3 Heading: gyro and GPS complementary filter

The guidance needs the *air* heading ψ (where the canopy points), not the ground track. Between GPS fixes the estimate is propagated by the bias-corrected gyro yaw rate r:

$$\hat{\psi} \leftarrow \hat{\psi} + r\,\Delta t. \tag{9}$$

At each GPS fix with ground speed $\geq 1.5\,\mathrm{m/s}$ the air velocity is formed by removing the wind estimate, and the heading is corrected with gain $k_\psi = 0.3$:

$$\psi_{meas} = \mathrm{atan2}\Big(v_E - \hat{W}_E,\; v_N - \hat{W}_N\Big) + r\,\tau_{lat}, \qquad \tau_{lat} = 0.15\,\mathrm{s}, \tag{10}$$

$$\hat{\psi} \leftarrow \hat{\psi} + k_\psi\ \mathrm{wrap}_{180}\Big(\psi_{meas} - \hat{\psi}\Big)\,. \tag{11}$$

The $r\,\tau_{lat}$ term compensates the GPS velocity delay during turns. Before a wind estimate exists, $\hat{\mathbf{W}} = 0$ and the filter tracks ground course, which makes HOMING steer the ground track at the target. The QMC5883 compass is used only when no GPS heading has arrived for 2 s, with gain 0.05, because it sits close to the servo motors and is not tilt-compensated.

5.4 Wind and airspeed from a circle fit

In steady flight the ground velocity is

$$\mathbf{v}_g = \mathbf{W} + V_a \begin{bmatrix}\cos\psi \\ \sin\psi\end{bmatrix}. \tag{12}$$

As the canopy turns, the points (v_N, v_E) lie on a circle whose **centre is the wind** and whose **radius is the airspeed**. No compass and no assumed airspeed are needed.

Algebraic least-squares fit. The last $N \leq 160$ GPS velocity samples (32 s at 5 Hz) are centred on their mean $(\bar{x}, \bar{y})$, giving x_i, y_i with $\sum x_i = \sum y_i = 0$. Fitting $x^2 + y^2 + ax + by + c = 0$ by minimising $\sum_i (z_i + ax_i + by_i + c)^2$ with $z_i = x_i^2 + y_i^2$ gives the normal equations

$$\begin{bmatrix} S_{xx} & S_{xy} \\ S_{xy} & S_{yy}\end{bmatrix}\begin{bmatrix} a \\ b\end{bmatrix} = -\begin{bmatrix} S_{xz} \\ S_{yz}\end{bmatrix}, \qquad c = -\frac{S_z}{N}, \tag{13}$$

with $S_{xx} = \sum x_i^2$, $S_{xz} = \sum x_i z_i$ and so on. Then

$$\hat{W}_N = \bar{x} - \tfrac{a}{2}, \qquad \hat{W}_E = \bar{y} - \tfrac{b}{2}, \qquad \hat{V}_a = \sqrt{\tfrac{a^2+b^2}{4} - c}. \tag{14}$$

Acceptance tests. A fit is rejected unless all of these hold:

- at least 30 samples, and $\det > 10^{-3} S_{xx} S_{yy}$ (samples not on a line);
- **turn coverage measured by the gyro**: the gyro-integrated heading of the samples must occupy at least 9 of 12 30° bins (270°) for the first fit and 7 bins (210°) afterwards, with at least 2 samples per bin;
- RMS radial residual $\leq 1.2\,\mathrm{m/s}$;
- $2.5 \leq \hat{V}_a \leq 15\,\mathrm{m/s}$ and $\|\hat{\mathbf{W}}\| \leq 12\,\mathrm{m/s}$;
- after the first fit, $|\hat{V}_a - \hat{V}_a^{prev}| \leq 2.5\,\mathrm{m/s}$.

Accepted fits are blended with weight 0.3. The fit is attempted once per second.

Why coverage comes from the gyro. The first version measured coverage as the angular spread of samples around the *fitted* centre. Monte-Carlo traces showed a short, noisy 60° arc being fitted by a small wrong circle, which then looked well covered around its own centre and produced a $5\,\mathrm{m/s}$ wind in still air. Measuring coverage from the independent gyro heading removed this failure.

Airspeed. When the wind is known and the canopy has flown straight for 2 s, $V_{a,str} = \|\mathbf{v}_g - \hat{\mathbf{W}}\|$ is low-pass filtered ($\tau = 4\,\mathrm{s}$). Priority: $V_{a,str}$, then the fit radius, then the default $6\,\mathrm{m/s}$.

6 Guidance Law

6.1 Virtual target

Define the time to go and the remaining air path

$$t_{go} = \frac{h}{V_z}, \qquad L = V_a\, t_{go}, \tag{15}$$

with V_z clamped to $[0.8, 8]\,\mathrm{m/s}$ and V_a to $[2.5, 15]\,\mathrm{m/s}$. The **virtual target** is

$$\mathbf{p}_{VT} = -\hat{\mathbf{W}}\, t_{go}. \tag{16}$$

Property: the virtual target is fixed in the air mass. Since $\dot{t}_{go} = -1$ (one second of flight removes one second of time to go),

$$\frac{d\mathbf{p}_{VT}}{dt} = -\hat{\mathbf{W}}\, \dot{t}_{go} = \hat{\mathbf{W}}. \tag{17}$$

The virtual target moves exactly with the wind, i.e. it is stationary relative to the air, and it reaches the real target at $t_{go} = 0$. Relative to the air, the canopy's motion is pure V_a flight with no wind. The whole windy problem therefore reduces to: *reach the air-fixed point* $\mathbf{p}_{VT}$ *using exactly* L *metres of air path.*

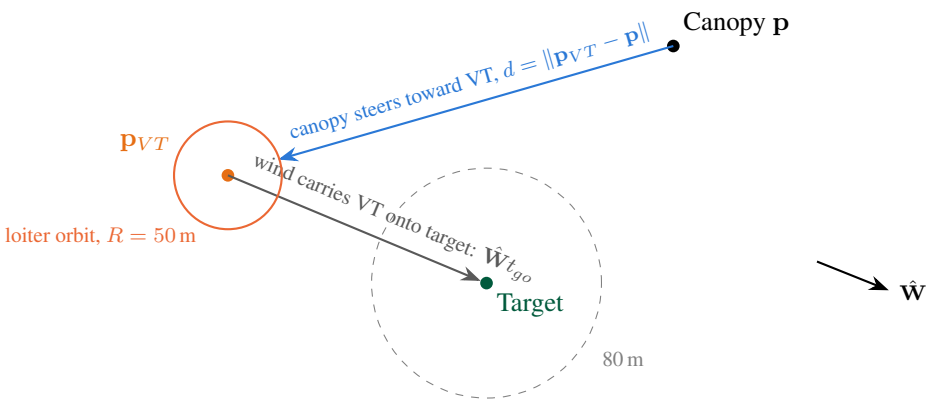


Figure 2: Geometry. The canopy steers toward the air-fixed virtual target; the wind delivers that point onto the real target at touchdown.

6.2 Excess energy

With $\mathbf{d} = \mathbf{p}_{VT} - \mathbf{p}$ and $d = \|\mathbf{d}\|$:

$$E = L - d \qquad [\mathrm{m}]. \tag{18}$$

$E > 0$: more glide than needed, which must be burned. $E < 0$: the target is out of reach (equivalent to Eq. (2) failing); the best action is to fly straight at $\mathbf{p}_{VT}$, which minimises the final distance because the touchdown point is $\mathbf{p} + \hat{\mathbf{W}} t_{go} + V_a t_{go}\,\hat{\mathbf{u}}$ and $\hat{\mathbf{u}} \parallel \mathbf{d}$ minimises its norm. The bearing to the virtual target is $\beta_{VT} = \mathrm{atan2}(d_E, d_N)$.

6.3 Phases and transitions

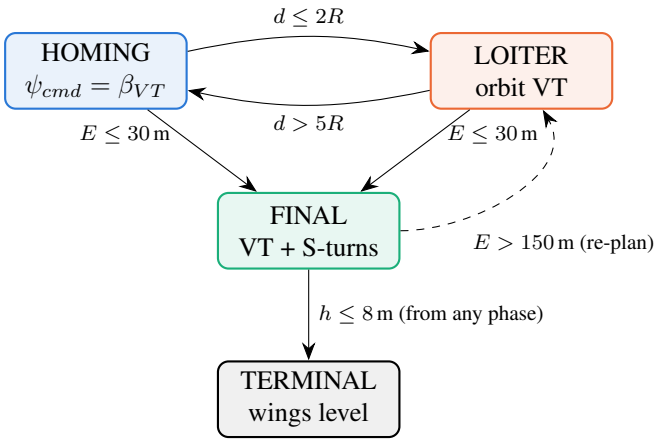


Figure 3: Guidance phases. From FINAL, a jump in excess energy above 150 m (for example after a sink-rate re-estimate) returns to LOITER if $d \le 2R$, otherwise to HOMING.

6.4 HOMING

$$\psi_{cmd} = \beta_{VT}, \qquad r_{ff} = 0. \tag{19}$$

6.5 LOITER: orbit vector field

Let γ be the bearing from the virtual target to the canopy, $R = 50$ m the orbit radius, $\lambda = +1$ for clockwise and $k_o = 1.5$. The commanded heading is

$$\psi_{cmd} = \gamma + \lambda \left[90^\circ + \arctan\left(k_o \, \frac{d - R}{R} \right) \right]. \tag{20}$$

On the circle ($d = R$) this is the tangent; far outside the correction tends to $+90^\circ$ and the command points at the centre; inside it points outward. A turn-rate feed-forward supplies the steady turn the orbit needs:

$$r_{ff} = \lambda \, \frac{V_a}{R} \cdot \frac{180}{\pi} \cdot \min\left(1, \frac{R}{d}\right) \quad [^\circ/\mathrm{s}]. \tag{21}$$

Because the circle is centred on an air-fixed point, it is a true circle in the air mass; over the ground it drifts downwind toward the target.

6.6 FINAL: S-turn energy burn

On entry the reference bearing $\psi_{ref} = \beta_{VT}$ is frozen. Flying at an angle δ from the line of sight consumes path L while closing distance d when $\cos\delta = d/L$, so

$$\delta = \begin{cases} \min\left(\arccos \dfrac{d}{L},\ 55^\circ \right) & E > 0, \\ 0 & E \le 0. \end{cases} \tag{22}$$

The side alternates on the cross-track error from the reference line,

$$x_t = (\mathbf{p} - \mathbf{p}_{VT}) \cdot \begin{bmatrix} -\sin\psi_{ref} \\ \cos\psi_{ref} \end{bmatrix} \quad \text{(positive right of the line)}, \tag{23}$$

with the sign s flipping from $+1$ to -1 when $x_t > 20\,\mathrm{m}$ and back when $x_t < -20\,\mathrm{m}$:

$$\psi_{cmd} = \beta_{VT} + s\,\delta. \tag{24}$$

Inside $12\,\mathrm{m}$ of the virtual target the command is held at ψ_{ref} so the canopy never pivots over the point near the ground.

6.7 TERMINAL

Below $h = 8\,\mathrm{m}$ steering is disabled and the servos slew to neutral. An optional symmetric flare at $3\,\mathrm{m}$ (both brakes pulled 350 counts) exists but is **disabled** until drop-tested.

6.8 Loss of GPS

If no valid fix arrives for $1.5\,\mathrm{s}$, the last ψ_{cmd} is held on the gyro-propagated heading for a further $8\,\mathrm{s}$. After that the canopy flies straight (neutral servos) until GPS returns.

6.9 Predicted miss (telemetry only)

$$\mathbf{p}_{land} = \mathbf{p} + \left(V_a \begin{bmatrix} \cos\hat{\psi} \\ \sin\hat{\psi} \end{bmatrix} + \hat{\mathbf{W}} \right) t_{go}, \qquad \text{predMiss} = \|\mathbf{p}_{land}\|. \tag{25}$$

6.10 Guidance update in pseudocode

Algorithm 1: `Guidance::update`, run at every GPS fix (5 Hz)

Input: $\mathbf{p}$, h, V_z, V_a, $\hat{\mathbf{W}}$, $\hat{\psi}$

$t_{go} \leftarrow h/V_z$; $\mathbf{p}_{VT} \leftarrow -\hat{\mathbf{W}} t_{go}$; $\mathbf{d} \leftarrow \mathbf{p}_{VT} - \mathbf{p}$; $L \leftarrow V_a t_{go}$; $E \leftarrow L - \|\mathbf{d}\|$;

if $h \le 8$ **then** phase $\leftarrow$ TERMINAL;

if *phase* = *HOMING* **then**

  if $E \le 30$ **then** enter FINAL ($\psi_{ref} \leftarrow \beta_{VT}$, $s \leftarrow +1$);

  else if $d \le 2R$ **then** phase $\leftarrow$ LOITER;

else if *phase* = *LOITER* **then**

  if $E \le 30$ **then** enter FINAL;

  else if $d > 5R$ **then** phase $\leftarrow$ HOMING;

else if *phase* = *FINAL* ***and*** $E > 150$ **then**

  phase $\leftarrow$ ($d \le 2R$? LOITER : HOMING);

compute ψ_{cmd}, r_{ff} for the phase (Sections 6.4 to 6.7);

Output: ψ_{cmd}, r_{ff}, steer flag, telemetry

7 Inner Loop: Turn-Rate Control

7.1 Control law

The heading error is converted into a turn-rate command, and the gyro closes a rate loop with feed-forward and integral action:

$$e = \mathrm{wrap}_{180}(\psi_{cmd} - \hat{\psi}), \qquad e \leftarrow 0 \text{ if } |e| < 2^\circ, \tag{26}$$

$$r_{cmd} = \mathrm{sat}_{\pm 22^\circ/s}(k_\psi\, e + r_{ff})\,, \tag{27}$$

$$I \leftarrow \mathrm{sat}_{\pm 250}(I + k_I\,(r_{cmd} - r)\,\Delta t)\,, \tag{28}$$

$$u^* = k_{ff}\, r_{cmd} + k_r\,(r_{cmd} - r) + I, \tag{29}$$

$$u^* \leftarrow u^* + \mathrm{sign}(u^*)\, u_{slack} \quad \text{if } |u^*| > 1, \tag{30}$$

$$u \leftarrow u + \mathrm{sat}_{\pm 900\,\Delta t}(\mathrm{sat}_{\pm 450}(u^*) - u)\,. \tag{31}$$

Defaults: $k_\psi = 0.6$ 1/s, $k_{ff} = 15$ counts per °/s, $k_r = 6$, $k_I = 8$, slack $u_{slack} = 0$ (to be measured), slew 900 counts/s. The integrator is frozen while the output is pinned at its limit in the same direction (anti-windup) and decays with $\tau = 1$ s when steering is disabled. The loop runs at 50 Hz.

7.2 Why the earlier law $u = K_p e - K_d r$ was replaced

An orbit requires a *steady* turn rate $r_{ss} = V_a/R$, which needs a steady brake input $u_{ss} = r_{ss}/G + u_{slack}$ where G is the canopy's turn rate per count. With $u = K_p e - K_d r$ the steady heading error is

$$e_{ss} = \frac{u_{ss} + K_d\, r_{ss}}{K_p}. \tag{32}$$

The damping term *fights* the turn the orbit needs. Example with $G = 0.04\,^\circ$/s per count, 80 counts of slack, $r_{ss} = 10\,^\circ$/s, $K_p = 5$, $K_d = 2.5$: $u_{ss} = 330$ and $e_{ss} = (330 + 25)/5 = 71^\circ$. The orbit cannot be held. The rate-command law removes the conflict: the feed-forward provides the expected input, and the integrator absorbs unknown gain and slack. The Monte-Carlo median miss fell from 24 m to 18 m when this change was made (Table 3).

7.3 Brake mapping

The canopy uses single-brake steering: only one line is pulled at a time.

$$u' = \text{TURN_SIGN} \cdot u, \qquad \begin{cases} u' > 0: & \text{right} = 2048 + s_R\,|u'|,\ \text{left} = 2048 \\ u' < 0: & \text{left} = 2048 + s_L\,|u'|,\ \text{right} = 2048 \end{cases} \tag{33}$$

with pull signs $s_L = s_R = -1$ (from the verified bench code) and positions clamped to $[200, 3895]$. Servo 1 is the left brake. A flare pulls both by 350 counts.

8 Flight-State Machine

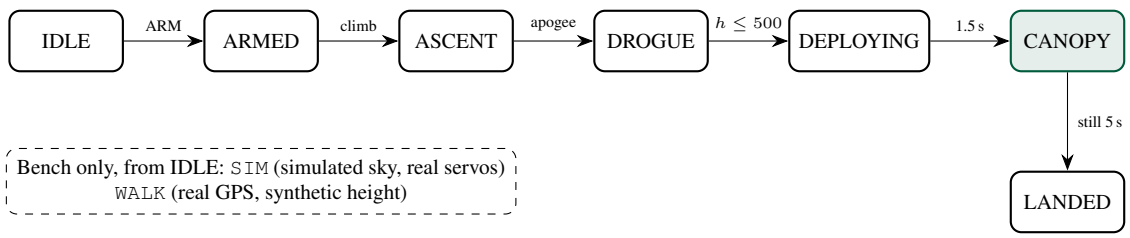


Figure 4: Flight-state machine in the `.ino`. Inside CANOPY the `FlightCore` runs DAMPING (8 s) → GUIDED → TERMINAL.

Transition	Condition
IDLE → ARMED	`ARM` command; both servos answer; baro or GPS altitude available. Captures p_0, gyro bias, GPS ground altitude; locks the SG90 at $140°$; opens the log.
ARMED → ASCENT	$h > 30\,\mathrm{m}$ and $\hat{v} > 8\,\mathrm{m/s}$ up
ASCENT → DROGUE	$\hat{v} < -2\,\mathrm{m/s}$ for $1\,\mathrm{s}$, or $h < h_{max} - 15\,\mathrm{m}$
DROGUE → DEPLOYING	$h \leq 500\,\mathrm{m}$ while descending; or, if $h_{max} < 520\,\mathrm{m}$, $2\,\mathrm{s}$ after apogee. SG90 to $180°$.
DEPLOYING → CANOPY	after $1.5\,\mathrm{s}$; `FlightCore.startDescent()`
CANOPY → LANDED	$h < 20\,\mathrm{m}$, $\lvert\hat{v}\rvert < 0.4\,\mathrm{m/s}$, ground speed $< 1\,\mathrm{m/s}$ for $5\,\mathrm{s}$; or $900\,\mathrm{s}$ timeout

Reset recovery. On entering ARMED, ASCENT, DROGUE, DEPLOYING or CANOPY the firmware writes a `ResumeRecord` (state, p_0, target, elevation offset, GPS ground altitude) to EEPROM, protected by an FNV-1a checksum. After a watchdog reset in flight it resumes: from DEPLOYING or CANOPY directly into guided flight, from ASCENT or DROGUE into DROGUE. The record is cleared on LANDED, IDLE or `CLEAR`.

9 Firmware Implementation

9.1 Scheduling

The main loop is fully non-blocking apart from short servo-bus pauses.

Task	Rate	Notes
GPS parse	every loop	one update per epoch (RMC trigger)
IMU yaw rate	100 Hz	heading propagation
Barometer	25 Hz	alpha-beta filter
Compass	10 Hz	fallback heading
Control (`FlightCore.step`)	50 Hz	inner loop
Servo write	20 Hz max	only on change ≥ 4 counts
USB telemetry `GUIDE`	4 Hz	switchable with `TLM`
SD log	10 Hz	flushed every 1 s

9.2 GPS configuration and quality gate

At boot a UBX-CFG-VALSET message (class 0x06, ID 0x8A, RAM layer) is sent:

Key	ID	Value	Verified
CFG-RATE-MEAS	0x30210001	200 ms (5 Hz)	**yes, 5.0 Hz measured**
CFG-NAVSPG-DYNMODEL	0x20110021	6 (airborne <1 g)	**not checked**
NMEA GSV / GLL / VTG on UART1	0x209100C5 / CA / B1	0 (off)	**not checked**

A fix is accepted when it is valid, younger than $1500\,\mathrm{ms}$, uses at least 5 satellites and has HDOP ≤ 4.0 (`FLIGHT`) or ≤ 8.0 (`LAB`). The earlier code's fixed limit of 3.5 had blocked all steering at HDOP 5.13 indoors.

9.3 BMP581 register driver

Soft reset (0xB6 to 0x7E), chip ID check (0x50, or 0x51 for BMP585), NVM status check and standby. Then OSR_CONFIG (0x36) is set to 0x58: pressure enabled, $8\times$ pressure and $1\times$ temperature oversampling.

ODR_CONFIG (0x37) is set to 0xBD: normal mode, 50 Hz, deep standby disabled. Six bytes from 0x1D give

$$T = \frac{\text{raw}_{24,\text{signed}}}{65536}\ [^\circ\text{C}], \qquad p = \frac{\text{raw}_{24}}{64}\ [\text{Pa}], \tag{34}$$

with a sanity window 30 kPa to 115 kPa.

9.4 Other sensors

- **QMC5883:** 0x0B ← 0x01, 0x09 ← 0x1D; calibrated (x, y) with offsets (180, 336) and scales (0.9565, 0.9044); navigation heading mirrored, $360^\circ - \text{atan2}(y, x)$, as found in the earlier compass test.
- **LSM6DSO32** (Adafruit LSM6DSOX driver): ±16 g, ±2000 °/s, 208 Hz. Yaw rate = $-\omega_z$ (Z axis up). Bias averaged over 200 samples at boot and at ARM.

9.5 ST-3009 servo bus

SCServo `SMS_STS` on Serial1 half-duplex at 1 Mbps. `WritePosEx(id, pos, 500, 50)`: speed 500 and acceleration 50 match the team's working stand-alone test. Each write checks the servo's acknowledgement; a missing ACK leaves the position unmarked so it is resent on the next 50 ms cycle. Torque is enabled explicitly at boot. A background PING every 2 s tracks bus health.

9.6 Command interface

Command	Action
`HELP`, `STATUS`	command list; full system status
`ARM`, `DISARM/STOP`, `CLEAR`	arm for flight; return to IDLE; erase resume record
`TARGET lat lon`, `TARGETHERE`, `ELEV m`	set target; target = current fix; target elevation offset
`DEPLOY`, `PREP`	SG90 release; SG90 lock
`CENTER`, `LEFT [n]`, `RIGHT [n]`, `MOTORTEST`	manual steering in IDLE
`PING`, `TORQUE`, `SPEED n`, `BUSTEST`	servo diagnostics (position, supply, temperature, missed ACKs)
`HEALTH ON|OFF`, `SERVODBG ON|OFF`	background PING; print every servo write
`SIM [n e h wN wE]`	simulated descent driving the real servos
`WALK`	walking test with real GPS, synthetic height
`GAINS kPsi kR kI kFF`, `SLACK n`, `LOITER R dir`	runtime tuning
`LAB`, `FLIGHT`	HDOP gate 8.0 or 4.0
`GYROTEST`, `BARO`, `LOG ON|OFF`, `TLM ON|OFF`	sensor checks, logging, telemetry

9.7 Telemetry record

Each `GUIDE` line (USB and SD) contains: time, flight state, core mode, phase, latitude, longitude, p_N, p_E, h, V_z, ground speed, $\hat{\psi}$, ψ_{cmd}, r_{cmd}, r, demand u, servo 1 and 2 positions, $\hat{W}_N$, $\hat{W}_E$, wind-valid flag, V_a, t_{go}, L, d, E, $\mathbf{p}_{VT}$, predicted miss, satellites, HDOP and GPS-OK flag. This record contains everything needed to measure the canopy's airspeed, sink rate and turn response after a drop test.

10 Monte-Carlo Verification

10.1 Method

The harness `host_sim/monte_carlo.cpp` runs complete descents through the *same* `FlightCore` object the Teensy runs. The controller is given only the defaults ($V_a = 6$, $V_z = 2.5\,\mathrm{m/s}$) and must measure everything else, as in flight.

10.2 Vehicle model

Brake slack u_0 is removed, the steady turn rate follows with a first-order lag, and turning increases sink:

$$u_{eff} = \mathrm{sign}(u)\max(|u| - u_0, 0), \qquad r_{ss} = \mathrm{sat}_{\pm r_{max}}(G\,\rho\,u_{eff})\,, \tag{35}$$

$$\dot{r} = \frac{r_{ss} - r}{\tau}, \qquad \dot{\psi} = r, \qquad V_z = V_{z0}\left[1 + \kappa\left(\frac{r}{r_{max}}\right)^2\right], \tag{36}$$

$$\dot{\mathbf{p}} = V_a\begin{bmatrix}\cos\psi\\ \sin\psi\end{bmatrix} + \mathbf{W}(h) + \mathbf{g}(t), \qquad \dot{h} = -V_z, \tag{37}$$

where ρ is the right/left gain ratio for right turns. Wind follows a power-law shear with directional veer, and gusts are a first-order Gauss-Markov process:

$$\mathbf{W}(h) = \left(\frac{h}{500}\right)^{\alpha}\mathbf{R}\big(\theta_v\,(1 - \tfrac{h}{500})\big)\,\mathbf{W}_{500}, \tag{38}$$

$$d\mathbf{g} = -\frac{\mathbf{g}}{\tau_g}\,dt + \sigma_g\sqrt{\frac{2\,dt}{\tau_g}}\,\mathbf{n}, \qquad \mathbf{n} \sim \mathcal{N}(0, I),\ \tau_g = 4\,\mathrm{s}. \tag{39}$$

10.3 Randomised parameters

Quantity	Distribution	Purpose
Airspeed V_a	$\mathcal{U}(5, 8)\,\mathrm{m/s}$	unknown canopy
Sink V_{z0}	$\mathcal{U}(1.8, 3.5)\,\mathrm{m/s}$	unknown canopy
Turn gain G	$\mathcal{U}(0.035, 0.09)\,^\circ/\mathrm{s}$ per count	unknown linkage
Right/left ratio ρ	$\mathcal{U}(0.8, 1.25)$	asymmetric rigging
Turn lag τ	$\mathcal{U}(0.3, 1.2)\,\mathrm{s}$	canopy inertia, pendulum
Brake slack u_0	$\mathcal{U}(0, 80)$ counts	line slack
Max turn rate	$\mathcal{U}(25, 45)\,^\circ/\mathrm{s}$	
Turn sink factor κ	$\mathcal{U}(0.2, 0.8)$	
Wind at 500 m	$\mathcal{U}(0, 6)\,\mathrm{m/s}$, any direction	
Shear exponent α, veer θ_v	$\mathcal{U}(0.08, 0.25)$, $\mathcal{U}(-25^\circ, 25^\circ)$	
Gust σ_g	$\mathcal{U}(0.2, 1.0)\,\mathrm{m/s}$	
Deployment height	$\mathcal{U}(470, 530)\,\mathrm{m}$	
Deployment distance	$\mathcal{U}(0, 900)\,\mathrm{m}$, any bearing	rocket drift
GPS	5 Hz, 0.15 s latency, 0.8 m white + random-walk position, 0.15 m/s velocity	
GPS dropout	20 % of flights, 10 s at a random time in 20 s to 120 s	
Baro	25 Hz, 0.25 m noise, ground bias $\mathcal{U}(\pm 3)\,\mathrm{m}$	
Gyro	$0.5\,^\circ/\mathrm{s}$ noise, residual bias $\mathcal{U}(\pm 0.6)\,^\circ/\mathrm{s}$	

Each flight uses its own random seed, so code changes are compared on identical flights.

10.4 Reachability classification

Using the true parameters, the reach margin at deployment is

$$M = V_a\,t_{av} - \left\|\mathbf{p}_0 + \bar{\mathbf{W}}\,t_{av}\right\|, \qquad t_{av} = \frac{h_0 - V_{z0}\,T_{damp}}{V_{z0}}, \tag{40}$$

with $\bar{\mathbf{W}}$ the wind averaged over the descent. $M < 0$: physically unreachable. $0 \le M \le 150\,\mathrm{m}$: marginal. $M > 150\,\mathrm{m}$: comfortable.

10.5 Results

Class (seed 99, 3000 flights)	N	≤80 m	Median	95th pct.
All flights (incl. unreachable)	3000	89.1 %	15.6 m	289 m
Reachable ($M \ge 0$)	2684	**98.3 %**	14 m	n/a
Marginal ($0 \le M \le 150$ m)	233	82.4 %	22.4 m	n/a
Comfortable ($M > 150$ m)	2451	**99.8 %**	13.9 m	34.4 m
Wind ≥ 4 m/s (all)	1007	n/a	22.6 m	n/a
GPS-dropout flights (all)	603	n/a	15.3 m	n/a

Table 1: Final configuration. Almost every miss above 80 m is a flight whose target was outside, or at the edge of, the canopy's physical reach.

Configuration	Reachable ≤80 m	Comfortable ≤80 m	Comfortable median / 95th
New, GPS 5 Hz (seed 99)	98.3 %	99.8 %	13.9 / 34.4 m
New, GPS 1 Hz (seed 99)	97.9 %	99.5 %	14.9 / 37.7 m
New, independent seed 123	98.2 %	99.8 %	13.7 / 32.5 m
Old point-at-target logic (seed 99)	97.2 %	98.9 %	29.4 / 61.8 m

Table 2: Comparisons. The old logic was emulated with loiter, final and wind compensation disabled (pure bearing pursuit, no 35 m capture).

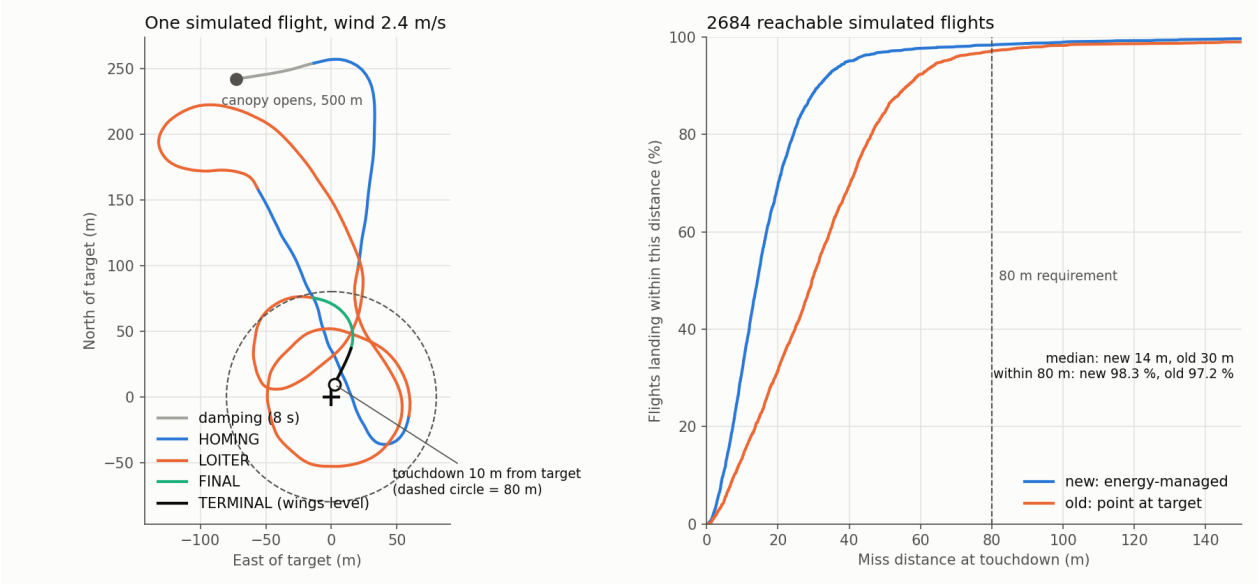


Figure 5: Left: one simulated flight through all phases (wind 2.4 m/s, touchdown 10 m from the target); the loiter circles drift downwind with the air mass. Right: cumulative miss distance over the reachable flights, new versus old logic.

An honest note on the old logic. Without its 35 m capture rule, pure pursuit already lands within 80 m in 97 % of reachable cases in this model, because overflying the target makes it spiral tightly overhead. The new law halves the median and 95th-percentile miss, but its larger real-world advantage is not captured by the

model: it flies gentle $50\,\mathrm{m}$ orbits at about $8\,^\circ/\mathrm{s}$ and keeps the wings level near the ground, whereas pursuit spirals at full brake for minutes, which on a real canopy risks a spiral dive, high sink and pendulum coupling.

10.6 Design iterations driven by the simulation

Version (seed 1, 1000 paired flights)	Reachable ≤80 m	Comfortable ≤80 m	Comfortable median / 95th
v1: open-loop wind-ID turn, $u = K_p e - K_d r$	92.5 %	97.2 %	23.8 / 58.9 m
v2: wind-fit coverage from gyro heading	92.0 %	97.7 %	23.7 / 54.5 m
v3: turn-rate inner loop + orbit feed-forward	93.1 %	98.5 %	18.0 / 40.8 m
v4: skip wind-ID turn when energy is tight	96.0 %	98.7 %	18.0 / 40.4 m

Table 3: Iteration history. v1 was run before per-flight seeding was added, so its flights are a different random draw; v2 to v4 are paired. v5 ($R = 50$ m, exit margin 30 m) and v6 (wind-ID turn removed) were evaluated on seed 99, Table 4.

Wind-ID turn policy (seed 99, 3000 flights)	Marginal ≤80 m	Comfortable ≤80 m
ID turn if reach margin > 300 m	50.6 %	99.1 %
ID turn if reach margin > 500 m	64.4 %	99.3 %
ID turn if reach margin > 800 m	77.3 %	99.6 %
No dedicated ID turn (final)	**82.4 %**	**99.8 %**

Table 4: A dedicated 330° wind-identification turn after damping always cost more altitude than it returned. The loiter turns measure the wind for free, so the ID turn is disabled by default.

10.7 Sensitivity (seed 99, 1500 flights, comfortable class)

Variation	≤80 m	Median	95th pct.
Baseline (baro ground bias ±3 m)	99.8 %	13.9 m	35.3 m
Baro / target-elevation bias ±10 m	99.7 %	18.8 m	45.0 m
Baro / target-elevation bias ±20 m	**98.6 %**	27.2 m	**63.7 m**
Gains k_ψ, k_r, k_I = 0.3, 2, 3	99.7 %	16.5 m	43.9 m
Gains = 0.4, 3, 4	99.8 %	14.9 m	38.9 m
Gains = 0.6, 6, 8 (default)	99.8 %	13.9 m	35.3 m
Gains = 0.8, 8, 10	99.8 %	13.4 m	33.6 m
R = 35 m, exit margin 30 m	99.8 %	14.9 m	38.7 m
R = 70 m, exit margin 30 m	99.8 %	15.2 m	36.8 m
R = 50 m, exit margin 15 / 45 / 60 m	99.8 %	16.6 / 17.9 / 20.7 m	40.8 / 36.7 / 41.8 m

The design is robust to gains and orbit size. The dominant external error is **height reference**: a wrong target-ground elevation directly scales t_{go}. The target elevation relative to the ARM point must be known within a few metres (`ELEV` command).

11 Hardware Bench and Ground Tests

Test	Result	Verdict
1. Compile and boot	IMU, BMP581 (chip 0x50), QMC5883 and both ST-3009 detected; gyro bias 0.32 °/s. SD and watchdog not yet installed.	pass
2. Steering servos	PING OK, positions 2048/2047, supply 7.6 and 7.7 V. `LEFT` moves only ID 1, `RIGHT` only ID 2.	pass
3. Gyro direction and scale	Clockwise rotation increases heading; a measured 180° turn read 180°.	pass
4. Barometer	p = 94 938 Pa, T = 22 °C; noise about ±0.15 m at rest; lift read +0.6 m (7.5 Pa) and returned to zero.	pass
5. SIM descent on real servos	`SIM 400 300 500 2 1`: full phase sequence, no missed ACKs, touchdown 13.9 m from target.	pass
6. GPS outdoors	12 satellites, HDOP 0.70, 5.0 Hz after the epoch fix.	pass
7. WALK test	Correct servo indicates turn toward the target; LOITER engaged within 100 m; GPS 5.0 Hz, HDOP 0.48.	pass
8. SD log and watchdog	pending	open

11.1 Defects found and fixed during bench testing

1. **Compile error "`ResumeRecord` does not name a type".** The Arduino IDE inserts function prototypes before the first function in the `.ino`, ahead of the struct. Fixed by moving the struct into `ResumeRecord.h`.

2. **Servos commanded but not moving, Servo 1 vibrating.** The team's stand-alone test moved both servos. Changes made: speed 2400 to 500 (matching the working test), explicit torque enable, and ACK checking with automatic resend. After these changes both servos respond correctly. *The single root cause was not isolated*; the ACK counter in `PING` should be watched in further tests.

3. **GPS rate reported as** 10 Hz**.** Each epoch produced two updates (GGA and RMC both carry position), so every fix was also fed to guidance twice. Fixed by triggering once per epoch on RMC.

12 Limitations and Open Items

- **Not yet flown or drop-tested.** Airspeed, sink rate, turn rate per count, lag and brake slack are modelling ranges. The Monte-Carlo results hold only if the real canopy lies inside those ranges.
- **Physical reach governs the requirement.** With $V_a = 6$ and $V_z = 2.5\,\mathrm{m/s}$ the still-air reach from 500 m is about 1.1 km after damping, less into a headwind. The deployment point relative to the target matters as much as the software.
- **Steering sign with the canopy attached.** `TURN_SIGN` and the pull signs are set from bench rotation only; a reversed sign would steer away from the target. It must be confirmed in the first drop.
- **Brake slack** (`SLACK`) is zero until measured.
- **Height reference.** `TARGET_ELEV_OFFSET_M` must be known within a few metres.
- **GPS settings** other than the measurement rate (dynamic model, NMEA trimming) are sent but not confirmed.
- **Flare** is disabled until tested.
- **Watchdog and SD logging** are not yet installed on the bench unit.

- **Model simplifications:** no pendulum dynamics, no canopy collapse, terrain assumed flat at target elevation.

13 Next Steps

1. Install the WDT_T4 library and a FAT32 microSD; verify `FLT000.CSV`.
2. **Drop test 1, steering sign:** `LEFT`/`RIGHT` fixed demands in turn, confirm the canopy turns in the commanded direction.
3. **Characterisation drops:** straight segments for V_a, V_z and glide ratio; step demands of 100 to 450 counts for turn gain G, lag τ and slack u_0, all from the `GUIDE` log.
4. Set `SLACK`, re-run the Monte-Carlo with the measured parameters narrowed in, and retune `GAINS` if needed.
5. Guided drops from increasing heights with a ground target; compare touchdown with the logged predicted miss.
6. Confirm the expected deployment point from the rocket trajectory lies well inside the canopy's reach envelope.

A Default Parameters

Parameter	Value	Where
Loiter radius R, direction	50 m, clockwise	`GuidanceConfig`
Orbit gain k_o	1.5	
Loiter entry / leave	$d \le 2R$ / $d > 5R$	
Final exit margin	30 m	
Final return-to-loiter	$E > 150$ m	
S-turn max angle / cross-track limit	55° / 20 m	
Final no-turn distance	12 m	
Terminal height	8 m	
Clamps: wind, V_z, V_a	12 m/s; 0.8 to 8 m/s; 2.5 to 15 m/s	
k_ψ, $r_{cmd,max}$	0.6 s^{-1}, 22 °/s	`SteeringConfig`
k_{ff}, k_r, k_I, I_{max}	15, 6, 8, 250 counts	
Heading deadband, max demand, slew	2°, 450 counts, 900 counts/s	
Damping time	8 s	`CoreConfig`
Wind-ID turn	disabled	
GPS timeout / heading hold	1.5 s / 8 s	
Heading GPS gain, latency comp.	0.30, 0.15 s	
Min speed for GPS heading	1.5 m/s (0.6 in WALK)	
Compass fallback gain	0.05	
Default V_a, V_z	6, 2.5 m/s	
Straight-flight threshold	$\lvert r\rvert < 6$°/s for 2 s	
τ_z, τ_{V_a}	3 s, 4 s	
Alpha-beta α, β	0.12, 0.008	
Wind window, solve period	160 samples, 1 s	
Parafoil deploy height	500 m	`.ino`
Launch detect	$h > 30$ m, $\hat{v} > 8$ m/s	
Apogee	$\hat{v} < -2$ m/s for 1 s or 15 m drop	

Parameter	Value	Where
Landed	$h < 20$ m, still 5 s	
GPS gate	$\geq$5 sats, HDOP $\leq$4 (8 in LAB), age $<$1.5 s	
Servo speed / accel	500 / 50	
Servo neutral / range	2048 / 200 to 3895	
Pull signs, `TURN_SIGN`	$-1, -1, +1$ (verify with canopy)	

B Reproducing the Monte-Carlo Results

```
cd Parafoil_Guided_Landing/host_sim
g++ -O2 -std=c++14 -o mc monte_carlo.cpp
./mc 3000 5 99            # final configuration, 5 Hz GPS
./mc 3000 1 99            # same flights, 1 Hz GPS
MC_OLD=1 ./mc 3000 5 99   # emulate the old point-at-target logic
MC_BARO=20 ./mc 1500 5 99 # +/-20 m height-reference error
MC_R=70 MC_EXTRA=30 ./mc 1500 5 99   # orbit radius / exit margin
MC_KPSI=0.8 MC_KR=8 MC_KI=10 ./mc 1500 5 99   # inner-loop gains
MC_IDMARGIN=300 ./mc 3000 5 99       # re-enable wind-ID turn
MC_FAIL=1 ./mc 1000 5 1   # print every flight that misses by > 80 m
```